CSD201 PE INSTRUCTIONS

Read the instructions below carefully before you start coding.

Students are ONLY allowed to use:

- Materials on his/her computer (including JDK, NetBeans...).
- For distance learning: Google Meet, Hangout (for Exam Monitoring Purpose).

Follow the steps below to complete PE:

1. Create a folder to store the given projects, e.g. CSD_given (1). Download the given materials into (1).
2. How to do question 1 (repeat the same for questions 2 and 3): Open NetBeans, open the given Q1 project, then edit the MyList.java file according to the requirements of the exam. (For Q2 and Q3, edit the files BSTree.java and Graph.java respectively).
3. Before submission: Run **"Clean and Build Project"** (Shift+F11) to make sure the build is SUCCESSFUL (otherwise the project will get 0 mark).
4. **Submission**: to submit project Q1, first select Question No = 1, browse and select the project folder (e.g. 1, Q1 or Q1X,...), then click the **Submit** button. Do the same for the other questions. **Do not submit** the un-edited given project. If the project is too big for submission, delete all f1.txt, f2.txt,....
5. **Do not use accented Vietnamese** when writing comments in programs.
6. **Do not add** new **import** statement(s) to given files.
7. Software tools must be used: **NetBeans IDE 8.x** and **Java JDK 1.8**. **If at least one of the above requirements is not followed, the exam will get ZERO.**

Trouble shooting:

If the given project (e.g. Q1) runs with an error, run "Clean and Build Project" (Shift+F11). If it still fails, try renaming or copying the project to another one, e.g. from Q1 to Q1X or Q1Y,...

Do not pay attention to the real meaning of objects, variables and their values in the questions below.

Question 1: (4 marks)

In this question, you need to complete some methods in the **MyList.java** file.

The class Bottle with 3 data members: maker, volume and color is given, and you do not need to edit it. The MyList class is a linked list of Bottle objects. Complete the following methods:

- **void addLast(String xMaker, int xVolume, int xColor)** - check if xMaker.charAt(0) == 'A' then **do nothing**, otherwise add new node with maker=xMaker, volume=xVolume, color=xColor to the end of the list. (volume and color can get arbitrary, even negative values).
- **void f1()** – Do not edit this method. You only need to complete the addLast(...) method above. Output in the file **f1.txt** must be the following:

```
(B,5,3) (C,6,5) (D,2,4) (E,7,9) (F,4,-7) (G,-3,2)
```

- **void f2()** – There are 3 given Bottle objects x, y, z in this function. Assume the list contains at least 3 elements. Write statements to insert x, y and z into the list so that x, y and z will be at positions 1, 2 and 4 (the head's position is 0). Output in the file **f2.txt** must be the following:

```
(C,9,8) (D,6,3) (E,8,5) (F,5,4) (I,4,9) (J,3,7)
(C,9,8) (X,1,2) (Y,2,3) (D,6,3) (Z,3,4) (E,8,5) (F,5,4) (I,4,9) (J,3,7)
```

- **void f3()** – Assume the list contains at least 3 elements. Remove the first node having the maximum color and insert it at position 1. (The head's position is 0). Output in the file **f3.txt** must be the following:

```
(C,7,6) (D,6,7) (E,3,8) (F,7,9) (G,8,7) (H,4,9) (I,8,3)
(C,7,6) (F,7,9) (D,6,7) (E,3,8) (G,8,7) (H,4,9) (I,8,3)
```

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- **void f4()** – Assume the list contains at least 6 elements and p is the 6th element in the list. Change the color of p to 88, then sort all elements before p in ascending order of color. The content of the output file **f4.txt** must be the following:

```
(C,9,8) (D,11,12) (E,8,7) (F,1,2) (I,7,9) (J,6,8) (K,5,6) (L,9,8) (M,3,4)
(F,1,2) (E,8,7) (C,9,8) (I,7,9) (D,11,12) (J,6,88) (K,5,6) (L,9,8) (M,3,4)
```

Question 2: (4 marks)

In this question, you need to complete some methods in the **BSTree.java** file.

The class Bottle with 3 data members: maker, volume and color is given, and you do not need to edit it. The BSTree class is a binary search tree of Bottle objects. The variable **volume is the key of the tree, thus it must be unique**. Complete the following methods:

- **void insert(String xMaker, int xVolume, int xColor)** - check if xMaker.charAt(0) == 'A' then **do nothing**, otherwise insert new Bottle object with maker=xMaker, volume=xVolume, color=xColor to the tree (volume and color can get arbitrary, even negative values).

- **void f1()** – Do not edit this method. You only need to complete the insert(...) method above. Output in the file **f1.txt** must be the following:

```
(B,9,4) (C,4,3) (D,8,6) (Y,6,-7) (E,2,5) (F,-6,7)
(F,-6,7) (E,2,5) (C,4,3) (Y,6,-7) (B,9,4) (D,8,6)
```

- **void f2()** – Perform a pre-order traversal from the root but write to file f2.txt only nodes with volume<7. **Hint**: Copy the function preOrder(...) to function preOrder2(...) and modify it. Output in the file **f2.txt** must be the following:

```
(C,8,2) (D,6,1) (F,2,-1) (I,1,7) (K,-1,5) (J,3,9) (L,5,10) (M,4,6) (G,7,3) (E,9,4) (H,10,8)
(D,6,1) (F,2,-1) (I,1,7) (K,-1,5) (J,3,9) (L,5,10) (M,4,6)
```

- **void f3()** – Assume p is the 6th node in the post-order traversal from the root. Delete node p by copying. Output in the file **f3.txt** must be the following:

```
(F,2,-1) (D,6,1) (G,7,3) (M,4,6) (K,-1,5) (I,1,7) (L,5,10) (J,3,9) (H,10,8) (E,9,4) (C,8,2)
(F,2,-1) (D,6,1) (G,7,3) (M,4,6) (K,-1,5) (L,5,10) (J,3,9) (H,10,8) (E,9,4) (C,8,2)
```

- **void f4()** – Assume p is the 3rd node having a right child in the pre-order traversal from the root. Count the number of nodes in the sub-tree with root p. If this number is k, change p.info.volume to 100+k. Output in the file **f4.txt** must be the following:

```
(C,8,2) (D,6,1) (F,2,-1) (I,1,7) (K,-1,5) (J,3,9) (L,5,10) (M,4,6) (G,7,3) (E,9,4) (H,10,8)
(C,8,2) (D,6,1) (F,2,-1) (I,1,7) (K,-1,5) (J,3,9) (L,5,10) (M,4,6) (G,7,3) (E,9,4) (H,106,8)
```

Question 3: (2 marks)

In this question, you need to complete some methods in the **Graph.java** file.

The class Graph implements a graph. Complete the following methods:

- **void f1()** – Perform a breadth-first traversal (to the file f1.txt) starting from vertex i=3 (vertex D), but display only 6 vertices, from the 3rd vertex to the 8th vertex. **Hint**: copy breadth(...) to breadth2(...) and modify the latter. The content of the output file **f1.txt** must be:

```
D A G B C E H I F
G B C E H I
```

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- **void f2()** – Apply Dijkstra's shortest path algorithm to find (1) the shortest path from vertex 1 (B) to vertex 5 (F), then (2) from vertex 0 (A) to vertex 6 (G). Write 3 lines to the file f2.txt: line 1 contains the vertices in the shortest path (1), line 2 contains the last 4 vertices selected into the set S in (2), line 3 contains the vertices with their labels in the shortest path (2). (Note that in the weighted matrix, the value 99 is considered as infinity). Output in the file **f2.txt** must be the following:

```
B C E D F
E D F G
A|0 B|10 C|12 E|15 D|19 G|29
```